NASA SPACE APPS CHALLENGE 2026

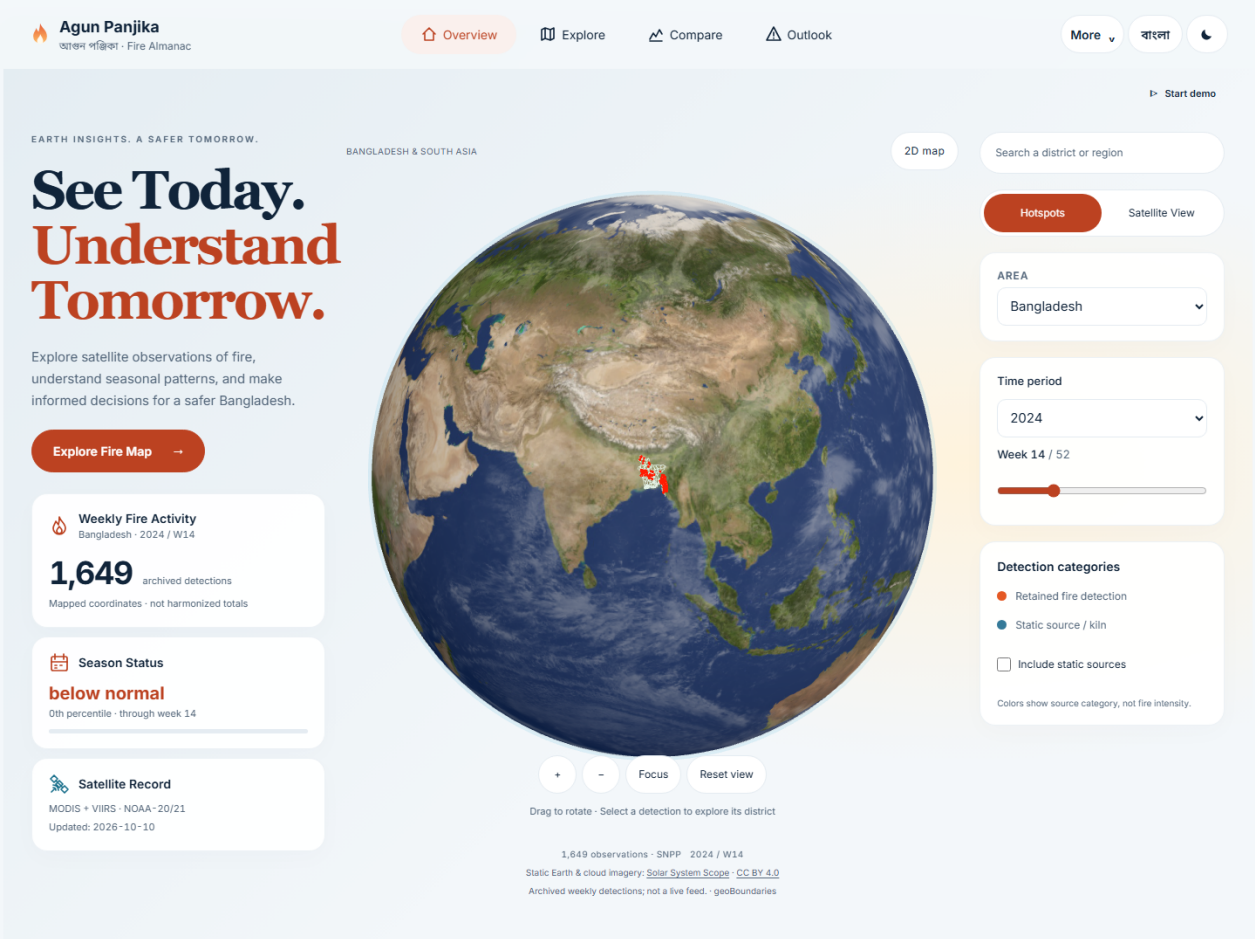
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Proposed global extension

Research priorities, features and evaluation milestones

From a Bangladesh prototype to independently evaluated fire calendars across selected regions.



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Proposal for mentor discussion
Prepared 29 October 2026. Extensions are future work; an award or globally valid accuracy is not guaranteed.

Source repository
Companion project brief
Detailed Markdown plan

1 Decision, scope and proposed contribution

Recommended direction

Make sensor comparability inspectable: a user chooses an evaluated area, compares its fire history across sensors, sees uncertainty and data sufficiency, and exports a dated evidence brief.

The current application already contains calendars, maps, a textured Earth, bilingual reports, guided storytelling and experimental outlooks. The strongest next contribution is better scientific and practical evidence for the central calendar, rather than more screens or visual effects.

What exists and what is proposed

Area	Current foundation	Proposed extension
Geography	Bangladesh, five regions, 64 districts; a Nepal configuration	Two or three independently evaluated, contrasting area packages.
Scientific comparison	Raw/factor/harmonized series and internal validation	Locked holdout, fold-local preprocessing/calibration and complete failure reporting.
Sensor explanation	Sensors screen and successor diagnostics	Held-out sensor-transition experiment linked to the calendar.
Interpretation	Uncertainty/source labels, bilingual reports	Visible data-sufficiency state and portable dated evidence brief.
Practical value	Intended users; an uncompleted study protocol	Recorded practitioner/comprehension pilot and revisions based on observed errors.

Table 1. Current capabilities are distinct from future deliverables. Country configuration alone does not establish transfer performance.

Three research questions

1. Does the reconstruction improve on strong simple baselines under independent temporal and geographic evaluation?
2. Are uncertainty and data-sufficiency statements reliable across activity levels, sensor changes and missing-data conditions?
3. Can intended users correctly interpret a season's status and limitations, and use the brief in a real workflow?

Scope boundary

The first extension should use **supported, evaluated areas**. Arbitrary polygon computation and operational warning integration follow only after suitable architecture and validation. No worldwide accuracy, agency adoption, smoke-arrival prediction or prevention benefit should be inferred from a global-looking map.

The proposal builds on the reviewed implementation and saved results. Its purpose is stronger validity, relevance and usefulness, not a promise of competition selection.

2 Work package 1: scientific independence

Lock the evaluation before extending the claims

The existing Bangladesh benchmark reports a 25.25% weekly-MAE improvement over a fitted factor. That result is internal. Strengthen the protocol before adding countries or repeating a broader accuracy claim.

1. **Reserve an untouched assessment.** Agree on temporal/geographic splits with a domain reviewer. Lock features, model weights, tuning and reporting rules before evaluating the final holdout.
2. **Fit learned steps within the split.** Static-source decisions, reconstruction/preprocessing and calibration must use permitted training data only. Evaluation-year exclusion must apply to every model fit used to generate calibration residuals, not only to the residual list.
3. **Separate reconstruction from forecasting.** The current reconstruction uses next-week MODIS counts. Define retrospective availability explicitly; prospective evaluation must respect issue-time source latency and use archived weather forecasts.
4. **Retain strong baselines.** Compare fixed 4.4×, fold-fitted and monthly factors, negative binomial and bagged LightGBM with the ensemble. Publish ablations and the conditions where a simpler model is adequate.

Evaluate uncertainty and failure modes

Report national/regional weekly MAE, off-season error, annual-total error, interval coverage and width. Include weak years, sparse areas and failures; average improvements alone are insufficient. Forecast intervals need separate calibration, and the observed-future-FWI experiment remains an optimistic sensitivity test.

Exercise sensor handover across main, district/local, neighbour and point products. Test real zeros, missing coverage, one/both successor sensors, month boundaries and a deliberately nonidentity correction. The current identity selection can conceal divergent product behavior.

For static-source filtering, lock the heuristic and inspect a new blind sample. Record positive, negative and uncertain labels, sampling uncertainty and wrongly removed vegetation fires. The tuned final 16/20 positive sample is not an independent precision/recall assessment.

Exit condition

A reviewer-agreed protocol, versioned split/input manifest and reproducible result table exist. Final results include failures and all stated baselines; uncertainty calibration is independently assessed. Passing thresholds are set before the final evaluation, not chosen after seeing its results.

Detailed source concerns: scientific review. Protocol and new experiments are proposed work; this report does not re-estimate the existing scores.

3 Work package 2: evaluated areas and transitions

A bounded global demonstration

Keep Bangladesh as the reference case, reproduce the Nepal example, and select a third setting with a domain reviewer because its fire seasonality, land cover or static-source mix tests a different assumption. A case is successful only if it meets predeclared criteria; unsuccessful cases remain visible as experimental or unsupported.

Each case should publish its boundary/version, sensor coverage and overlap, training/evaluation cutoffs, baselines, the same metric definitions, interval diagnostics and failure analysis. Any independent burned-area/local evidence is described as a cross-check of its own observable. Fresh reproducible results are needed before using an older Nepal claim in a new submission.

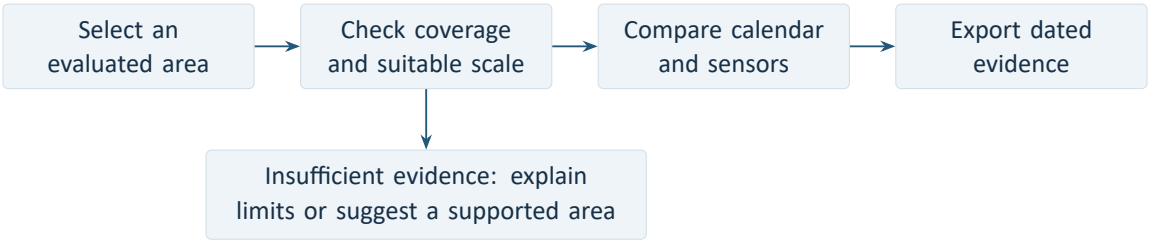


Figure 1. Proposed interaction. Abstention is part of the scientific contract, rather than a missing feature disguised as zero activity.

Area packages before arbitrary polygons

The current model learns region-week aggregates. A polygon-drawing control cannot create a calibrated history by filtering the existing point export: required features, suitable overlap and scale-dependent evaluation are missing. Begin with prebuilt area packages that the static app can select.

An area specification should contain geometry/boundary version, time window, source sensors, coverage thresholds and data/model version. The pipeline produces a reproducible evaluated package, preserving observed/estimated provenance. Later custom areas need asynchronous/server computation or a validated spatial aggregation method; keep FIRMS credentials on the pipeline/server side.

Make a sensor transition inspectable

Extend the existing Sensors screen with one area, period and transition date. Display raw observations, naive stitching, a fitted-factor result and harmonized history on comparable axes. In an overlap experiment, withhold the reference sensor for a period, reconstruct it and reveal the held-out observations. Expose the training/evaluation periods.

The accompanying evidence panel and export should retain area, source, completeness, model version, error, interval and data-through date. For periods without a reference sensor, label reconstruction and uncertainty explicitly.

Exit condition: an added area is reproducible and either meets its declared requirements or visibly abstains. A user can distinguish the sensor artifact, the supported comparison and the estimated period.

4 Work package 3: users, access and release

Test comprehension and workflow fit

Use the existing five-person protocol as a pilot. Include at least one relevant practitioner if feasible and arrange native Bangla/CHT terminology review. Intended users and personas do not substitute for participants or evidence of adoption.

Ask each participant to choose an area, judge whether a selected season is unusual, identify estimated periods/uncertainty, and share a report that preserves its caveats. Record correctness, time, language, device and the participant's explanation; revise the interface from observed misunderstandings.

Pilot task	Proposed acceptance evidence
Interpret a season	At least four of five correct answers; median time below ten seconds, following the repository's pilot targets.
Understand provenance	At least four of five can find the data date and explain estimated/observed periods and the interval.
Recognize insufficient data	Participants distinguish missing/sparse coverage from low activity and explain the abstention.
Fit an actual workflow	A practitioner records the decision, objections and evidence needed; review is not agency adoption.
Use it on a phone/offline	Keyboard/table access and readable small-screen report; cached resources remain usable with disclosed limits.

Table 2. Pilot criteria are proposed, not achieved outcomes. A larger study is needed for population-wide usability or impact claims.

Measure the existing sub-three-second mobile LCP target on a declared device/throttling setup. Local desktop timings do not establish that target. Do not infer fires prevented, lives saved or operational warning benefit from a small comprehension study.

Make the result reliably reviewable

- Diagnose the hosted pipeline failure, publish a working demo URL and verify refresh/build/deploy. A local pass does not establish hosted operation.
- Expose source completeness, version, data-through date and stale status. Expire or clearly label old forecast products; missing acquisition chunks must not silently appear as zero activity.
- Synchronize Method text, translations, citations, README and video claims with one result table. The current in-app 50/50 and 2012–2024 wording differs from the 30/70, 2012–2025 model/export.
- Review dependency advisories separately, choose an explicit software license, and preserve data/imagery credits and required AI-use disclosures.

Exit condition: a dated public demo works, a pilot is completed, misunderstandings produce documented revisions, and a portable evidence brief preserves the scientific limits.

5 Milestones, dependencies and risks

The sequence below is dependency-based. It is not a set of competition deadlines or an existing team commitment. Assign owners and timing after the mentor helps narrow scope.

Milestone	Deliverable	Exit condition
A. Baseline	Aligned method/result note, frozen snapshot and public demo	Link works; freshness/version visible; local checks and hosted refresh pass.
B. Protocol	Agreed protocol and reproducible benchmark	Splits/calibration reviewed; final scores and failures reported.
C. Transfer	Additional evaluated area and transition comparison	Reproducible package with baseline/error/coverage evidence and support limits.
D. Pilot	Completed pilot and revised user journey	Task/uncertainty comprehension recorded; revisions trace to observed errors.
E. Package	Short demo/slides, description, evidence links and credits	Applicable rules confirmed; public links work; numerical claims trace to artifacts.

Table 3. Complete one bounded deliverable before broadening geography, architecture or operational claims.

Research and engineering risks

Risk	Response in the plan
Transfer fails or requires recalibration	Publish the failure and local adjustments; limit supported cases rather than claim global validity.
Coverage/calibration is unreliable	Set requirements with a reviewer, show interval width and abstain on unsupported conditions.
Custom areas exceed the static architecture	Deliver evaluated area packages first; select new processing infrastructure from measured needs.
Feature growth outpaces evidence	Prioritize the benchmark, one transition workflow and one user journey.
Rules or permitted edits are uncertain	Confirm with organizers before treating later work as competition-evaluable.

Smallest credible next deliverable

One reproducible, independently assessed comparison beyond the baseline, linked to a working calendar and a user who can explain its uncertainty. This can be a stronger Bangladesh case or a validated transfer case, depending on the mentor's advice.

6 Submission story and mentor decisions

A focused demonstration

Show the sensor discontinuity, reveal the comparable calendar and uncertainty, demonstrate one evaluated transfer case, and close with the user's decision. Use only metrics that can be explained and reproduced. The current 2:42 video remains useful background; a separate short global demonstration may be necessary.

The supplied Bangladesh guide distinguishes local 240-second presentations from a global under-30-second video or seven-slide alternative. It calls for public links, English presentation/subtitles, source credits and AI-use disclosure. It contains conflicting cutoffs and is not independently established as final global policy. Confirm format, deadlines and permitted edits with the Local Lead and final official guidance.

What a stronger submission would show

Review lens	Inspectable evidence
Scientific validity	Independent evaluation, baseline comparisons, calibrated uncertainty and complete failure reporting.
Challenge relevance	A comparable history and calendar through an actual or held-out sensor transition.
Practical impact	Recorded interpretation tasks and a practitioner's documented workflow need.
Creativity	An understandable transition reveal and a useful insufficient-evidence state.
Presentation	Working public materials, readable dated brief, focused demo and source attribution.

Table 4. Planning lenses, not a confirmed global scoring formula or a prediction of an award.

Four decisions to take to the mentor

1. What independent evaluation is necessary for the central harmonization claim?
2. Is the next priority a stronger Bangladesh pilot or an evaluated additional-country case?
3. Which user decision, practitioner and domain reviewer should shape the prototype?
4. How should the contribution be positioned against existing sensor-fusion work?

Supporting material

- Project brief, scientific review, verification record, user-test protocol and credits/submission notes.
- Supplied 2026 Bangladesh Participants Guide; official Space Apps site; team video.

NotebookLM required login and the supplied Claude artifact content was inaccessible during the review; neither was used to infer rules. The detailed Markdown plan retains the fuller evidence and source-access notes.